Vathsal Vuppala

linkedin.com/in/vathsalvuppala | github.com/vathsal14

vathsal14@gmail.com | +91 87904 23358
Hyderabad, India

Professional Summary

AI Software Developer with 1.5+ years of experience designing and shipping production-grade intelligent systems. Specializes in LLM-based multi-agent architectures, agentic workflow design (LLaMA, Mistral, LangChain), and end-to-end full-stack AI platforms — from model training and inference APIs (Python, FastAPI) to front-end interfaces (React). Experienced across clinical AI and ed-tech domains, with a focus on systems that are reliable, maintainable, and built for scale.

Education

- **CVR College of Engineering (JNTUH)** — Hyderabad, Telangana
  *B.Tech in Computer Science and Information Technology | CGPA: **8.7 / 10*** — *July 2021 – May 2025*

Experience

- **SAAP Technologies** — Hyderabad, Telangana
  AI Software Developer — *Dec 2024 – Present*
  - **NOVO Wellness — Student Mental Health AI Platform**: Architected an end-to-end agentic AI platform for students, teachers, and schools using Python, FastAPI, and React. Designed multi-agent pipelines with LLaMA and Mistral for psychometric assessment interpretation, student profiling, and personalized recommendations. Dashboards enabled early identification of behavioral patterns and improved counselor intervention speed.
  - **NOVO Neurotech — Clinical Neurological Insights System**: Built AI analysis pipelines for neurological scan data to support doctor decision-making. Implemented DICOM preprocessing, feature extraction, and deep learning model training using TensorFlow and PyTorch. Deployed scalable inference APIs for consistent clinical interpretation across multi-site environments.

Projects

- **PPE Safety Compliance Detection System**
  Python, YOLOv8, DeepSORT, FastAPI, React, PostgreSQL
  Real-time workplace safety monitor using YOLOv8 to detect missing PPE (helmets, vests, gloves) from live video feeds with DeepSORT worker tracking, a FastAPI inference backend, and a React dashboard with live violation alerts and compliance logs.

- **Autonomous Barista Agent**
  Python, LangChain, OpenAI API, Twilio Voice API
  Voice-enabled LLM agent that accepts live customer calls, interprets natural language orders, and uses tool calling to write and confirm orders in real time — with persistent memory, fallback logic, and multi-step orchestration for modifications and edge cases.

- **Agentic Seminar Report Writer**
  Python, LangChain, LlamaIndex, ChromaDB, FastAPI, React
  Multi-agent RAG platform that takes a seminar topic, retrieves and synthesises research papers via vector search, then autonomously plans, writes, and critic-reviews a structured technical report with IEEE citations and PDF/DOCX export via a React frontend.

Certifications

- **Deep Learning Specialization**: Coursera / DeepLearning.AI — Neural networks, CNNs, RNNs, optimization (Andrew Ng)
- **Machine Learning with Python**: IBM / Coursera — Supervised & unsupervised learning with Scikit-learn
- **Generative AI with LLMs**: Coursera / DeepLearning.AI & AWS — Fine-tuning (LoRA/PEFT), RLHF, LLM deployment

Skills

- **Languages**: Python, JavaScript (ES6+), TypeScript, Java, C, HTML5, CSS3
- **AI / ML & LLMs**: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LangChain, LlamaIndex, LLaMA, Mistral, OpenAI API, Hugging Face Transformers, PEFT / LoRA, RAG, Prompt Engineering, CNNs, RNNs, DQN
- **Agentic & Infra**: Multi-Agent Workflows, Tool Use / Function Calling, Vector Databases (FAISS, ChromaDB), FastAPI, REST APIs, Docker, Linux, Git
- **Full-Stack**: React, Next.js, Node.js, Express.js, Flask, Django, Tailwind CSS
- **Databases & Cloud**: MySQL, MongoDB, PostgreSQL, Firebase, Azure (App Service, Blob Storage)